

Section : Common

Q.1

The degree of the differential equation $\left(2 + \left(\frac{dy}{dx}\right)^2\right)^{\frac{3}{2}} = a^2 \frac{d^2y}{dx^2}$ is:

1. 6
2. 1
3. 3
4. 2

Options 1. 1

2. 2
3. 3
4. 4

Q.2

If A is a square matrix of order 3 and $|A| = -5$, then $|3A|$ is equal to

1. -135
2. -15
3. 135
4. 45

Options 1. 1

2. 2
3. 3
4. 4

Q.3

If A is an invertible symmetric matrix, then A^{-1} is

1. a symmetric matrix
2. a skew-symmetric matrix
3. neither a symmetric matrix nor a skew-symmetric matrix
4. always an identity matrix

Options 1. 1

2. 2
3. 3
4. 4

Q.4

If $y = (\log x)^{\log x}$, $x > 1$ then $\frac{dy}{dx}$ is equal to

1. $\log x \left[\frac{1 + \log(\log x)}{x} \right]$, $x > 1$
2. $(\log x)^{\log x} \left[\frac{1 + \log x}{x} \right]$, $x > 1$
3. $(\log x)^{\log x} \left[\frac{x + \log(\log x)}{x} \right]$, $x > 1$
4. $(\log x)^{\log x} \left[\frac{1 + \log(\log x)}{x} \right]$, $x > 1$

Options 1. 1

2. 2
3. 3
4. 4

Q.5

The random variable X has a probability distribution $P(X = r) = \begin{cases} rk, & \text{if } r \leq 2, \\ (r-1)k, & \text{if } 2 < r \leq 4, \text{ where } r \in \mathbb{N} \cup \{0\} \text{ and } k \in \mathbb{R}, \text{ where } n \text{ is the set of natural numbers} \\ 0, & \text{otherwise,} \end{cases}$

- (A) $k = \frac{1}{9}$
- (B) $P(2 \leq X \leq 3) = \frac{1}{2}$
- (C) $P(X = 4) = \frac{1}{3}$
- (D) $P(X > 1) = \frac{7}{8}$

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (A), (C) and (D) only
3. (B) and (D) only
4. (A) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.6

The value of $\int_0^1 [\log x - \log(1-x)] dx$ is

1. $\frac{1}{2}$
2. 2
3. 1
4. 0

Options 1. 1

2. 2
3. 3
4. 4

Q.7

The area of the region bounded by $y^2 = 9x$, $x = 2$, $x = 4$ and the x -axis in the first quadrant, is

1. $16 - \sqrt{2}$ sq. units
2. $16 + \sqrt{2}$ sq. units
3. $16 + 4\sqrt{2}$ sq. units
4. $16 - 4\sqrt{2}$ sq. units

Options 1. 1

2. 2
3. 3
4. 4

Q.8

The corner points of the bounded feasible region determined by a set of constraints (linear inequalities) are $A(0, 5)$, $B(3, 5)$, $C(5, 0)$, $D(4, 1)$ and the objective function is $z = px + 2qy$ where $p, q > 0$.

The condition on p and q such that the maximum z occurs at B and D, is

1. $p - 8q = 0$
2. $q - 5p = 0$
3. $p - 5q = 0$
4. $p - 3q = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Match List-I with List-II

Consider the function $f(x) = 2x^3 - 21x^2 + 36x + 80$, $x \in [0, 6]$.

List-I**List-II**

- (A) one of its critical points is at $x =$ (I) -28
 (B) Its absolute maximum value is (II) -42
 (C) Its absolute minimum value is (III) 97
 (D) Its second derivative at $x = 0$ (IV) 6

Choose the correct answer from the options given below:

1. (A) $- (I)$, (B) $- (III)$, (C) $- (II)$, (D) $- (IV)$
2. (A) $- (III)$, (B) $- (IV)$, (C) $- (II)$, (D) $- (I)$
3. (A) $- (IV)$, (B) $- (III)$, (C) $- (I)$, (D) $- (II)$
4. (A) $- (IV)$, (B) $- (II)$, (C) $- (III)$, (D) $- (I)$

Options 1. 1

2. 2
3. 3
4. 4

Q.10

If a matrix has 12 elements, then the possible orders it can have, are

- (A) 1×12
- (B) 4×3
- (C) 6×3
- (D) 6×2

Choose the correct answer from the options given below:

1. (A), (B) and (D) only

2. (A), (B) and (C) only
3. (B), (C) and (D) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.11 If $A^T = \begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}$, then the matrix $(A + 2B)^T$ is

1. $\begin{bmatrix} 4 & 5 \\ -1 & -6 \end{bmatrix}$
2. $\begin{bmatrix} -4 & -5 \\ 1 & 6 \end{bmatrix}$
3. $\begin{bmatrix} -4 & 5 \\ 1 & 6 \end{bmatrix}$
4. $\begin{bmatrix} 4 & -5 \\ -1 & 6 \end{bmatrix}$

Options 1. 1

2. 2
3. 3
4. 4

Q.12 $\int \frac{(x-3)e^x}{(x-1)^3} dx$ is equal to

1. $\frac{e^x}{(x-1)} + C$, C is an arbitrary constant
2. $\frac{e^x}{(x-3)^2} + C$, C is an arbitrary constant
3. $\frac{e^x}{(x-1)^2} + C$, C is an arbitrary constant
4. $\frac{e^x}{(x-1)^3} + C$, C is an arbitrary constant

Options 1. 1

2. 2
3. 3
4. 4

Q.13 If the interval in which $f(x) = \frac{x}{4} + \frac{4}{x}$, $x \neq 0$ is strictly increasing is $(-\infty, a) \cup (b, \infty)$, then

1. $a = b$
2. $a = -b$
3. $a = 2b$
4. $a = -2b$

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Consider the following L.P.P.

Minimize $z = 400x + 300y$, subject to $100x + 200y \geq 12000$, $300x + 400y \geq 20000$, $200x + 100y \geq 15000$

and $x, y \geq 0$. Then

1. its feasible region is bounded and optimal point is (60, 30)
2. its feasible region is unbounded and optimal point is (60, 30)
3. its feasible region is bounded and optimal point is (0, 110)
4. its feasible region is unbounded and optimal point is (30, 70)

Options 1. 1

2. 2
3. 3
4. 4

Q.15

The particular solution of the differential equation $\log \left(\frac{dy}{dx} \right) = 3x + 4y$ satisfying $y = 0$ when $x = 0$ is:

1. $3e^{3x} + 4e^{-4y} + 7 = 0$
2. $3e^{3x} - 4e^{-4y} - 7 = 0$

$$3. 4e^{3x} + 3e^{-4y} - 7 = 0$$

$$4. 4e^{3x} - 3e^{-4y} + 7 = 0$$

- Options
1. 1
 2. 2
 3. 3
 4. 4

Section : Applied Mathematics

Q.16 The point on the curve $y = (x - 2)^2$ at which the tangent is parallel to the chord joining the points (2, 0) and (4, 4) is:

1. (3, -1)
2. (2, 3)
3. (2, -3)
4. (3, 1)

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.17 A boat covers 32 km upstream and 36 km downstream in 7 hours. Also, it covers 40 km upstream and 48 km downstream in 9 hours. The speed of the boat in still water is:

1. 12 km/hr
2. 8 km/hr
3. 9 km/hr
4. 10 km/hr

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.18 Consider the linear programming problem (LPP):

$$\text{Maximize } Z = 6x + 3y$$

subject to the conditions

$$4x + y \geq 80$$

$$x + 5y \geq 115$$

$$3x + 2y \leq 150$$

$$x \geq 0, y \geq 0$$

In reference to the above LPP, which of the following are correct?

- (A) The feasible region is bounded.
- (B) The corner points of the feasible region are (15, 20), (40, 15) and (0, 75).
- (C) The maximum value of the objective function is 285.
- (D) The LPP does not have optimal solution.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A) and (C) only
4. (B) and (D) only

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.19 The following data shows the percentage of rural and urban Indian households who have a high speed internet connection.

Year(x) Rural Household (y%) Urban Household(y%)

2020	2	6
2021	3	7
2022	3	9
2023	4	15
2024	5	20

Based on the above data, which of the following statements are correct?

- (A) A straight line trend by the method of least squares for rural Indians is $y_t = 3.4 + 0.7(x - 2022)$
- (B) A straight line trend by the method of least squares for urban Indians is $y_t = 11.4 + 4.3(x - 2022)$
- (C) The forecast for the year 2025 for urban group using the trend equation is 22.2%
- (D) The forecast for the year 2025 for rural group using the trend equation is 5.5%

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (A) and (D) only
4. (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.20 In a game, a man wins ₹ 8 for getting a number greater than 3 and loses ₹ 3 otherwise, when a fair die is thrown.

The man decided to throw a die 4 times but to quit as and when he gets a number greater than 3.

If X denotes the amount which the man wins or loses, then which of the following are correct?

(A) All the possible values of X are 8, 5, 2 and -1 .

(B) The probability distribution of X is:

X 8 5 2 -1 -12

$P(X)$ $\frac{1}{2}$ $\frac{1}{4}$ $\frac{1}{8}$ $\frac{1}{16}$ $\frac{1}{16}$

(C) The mean value of X is $\frac{75}{16}$.

(D) The variance of X is $\frac{6615}{256}$.

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A), (B) and (C) only
3. (B), (C) and (D) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.21 Shyam invested ₹ 2,00,000 in 2019 for 5 years.

If the compound annual growth rate (CAGR) for his investment is 10%, then the end balance of his investment is:

1. ₹ 3,22,102
2. ₹ 3,49,900
3. ₹ 3,49,960
4. ₹ 3,60,490

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Which of the following are normal equations to fit a straight line trend $y = a + bx$ by the method of least squares?

1. $\sum y = a \sum x + b \sum x^2$, $\sum xy = a \sum x + b \sum y^2$
2. $\sum y = na + b \sum x$, $\sum xy = a \sum x + b \sum x^2$
3. $\sum xy = na + b \sum x$, $\sum xy = na + b \sum x^2$
4. $\sum x = na + b \sum y^2$, $\sum y = a \sum x + b \sum y^2$

Options 1. 1

2. 2
3. 3
4. 4

Q.23 A machine has been producing steel tubes of mean inner diameter of 2 cm.

A sample of 10 tubes gives an inner diameter of 2.01 cm

and a standard deviation of 0.063 cm.

Test the hypothesis that the machine is working in the proper order at 5% level of significance and answer, which of the following statements are correct?

[Given that $t_{0.05} = 2.262$]

- (A) The value of the test statistic is $t = 0.476$
- (B) The null hypothesis is rejected at 5% level of significance.
- (C) The null hypothesis is accepted at 5% level of significance.
- (D) The degree of freedom is 10

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (A) and (C) only

4. (C) and (D) only

- Options 1. 1
2. 2
3. 3
4. 4

Q.24 If the random variable X is normally distributed with mean 16 and standard deviation 4, then the standard normal variable Z corresponding to $X = 17$ is

1. 1.25
2. -1.25
3. -0.25
4. 0.25

- Options 1. 1
2. 2
3. 3
4. 4

Q.25 The fund created to accumulate money over the years to discharge a future obligation is known as:

1. Emergency Fund
2. Perpetuity
3. Sinking Fund
4. Equity

- Options 1. 1
2. 2
3. 3
4. 4

Q.26 In an influenza epidemic, the number of cases diagnosed in a week were as follows:

Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Number of Cases	10	12	8	16	9	14	4

The 3-day moving averages are:

1. 10, 11, 12, 13, 9
2. 10, 11, 13, 12, 9
3. 10, 12, 14, 13, 11
4. 10, 12, 11, 13, 9

- Options 1. 1
2. 2
3. 3
4. 4

Q.27 The value of $\begin{bmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(c+a) \\ 1 & ab & ab(a+b) \end{bmatrix}$ is

1. 1
2. 0
3. abc
4. $(a+b+c)^2 abc$

- Options 1. 1
2. 2
3. 3
4. 4

Q.28 The corner points of the bounded feasible region for a linear programming problem (LPP) are $(0, \frac{3}{2})$, $(1, 2)$ and $(4, 0)$.

If the objective function is $Z = ax + by$, where a and b are positive, then the condition on a and b so that the maximum of Z occurs at $(1, 2)$ and $(4, 0)$ is:

1. $a = 2b$
2. $2a = b$
3. $3a = 2b$
4. $3a = 4b$

- Options 1. 1
2. 2
3. 3
4. 4

Q.29 If a moneylender charges interest at the rate of 10 rupees per 100 rupees per half year, payable in advance, then the effective rate of interest per annum is:

1. 10%
2. 11.1%
3. 25%
4. 23.45%

Options

1. 1
2. 2
3. 3
4. 4

Q.30 If A is a square matrix of order 3 and $|A| = 4$, then the value of $|2A^T|$ is

1. 8
2. -8
3. -32
4. 32

Options

1. 1
2. 2
3. 3
4. 4

Q.31 Which of the following are correct?

- (A) For a square matrix A , if $A^3 = I$, then $A^{-1} = A^2$.
(B) The determinant of only a square matrix can be defined.
(C) If A is a square matrix of order 3, then the number of minors of the matrix A is 3.
(D) If A and B are two non-singular matrices of the same order, then $(AB)^{-1} = B^{-1}A^{-1}$.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (B) and (D) only
4. (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.32 Mr. X purchased a house for ₹ 7,00,000 and made a down payment of ₹ 1,50,000.

He repays the balance in 25 years by equal monthly installments at 9% per annum compounded monthly.

The equated monthly installment (EMI) is:

[Given that $(1.0075)^{-300} = 0.106$]

1. ₹ 3625
2. ₹ 4614
3. ₹ 5614
4. ₹ 4714

Options

1. 1
2. 2
3. 3
4. 4

Q.33 Match List-I with List-II

List-I

List-II

(Parametric Equations) $\frac{dy}{dx}$

(A) $x = \frac{2}{t}, y = 2t$ (I) $4t^2$

(B) $x = t^3, y = 3t + 2$ (II) $2(t + 1)$

(C) $x = \log t, y = 2t^2$ (III) $-t^2$

(D) $x = e^t, y = 2te^t$ (IV) t^{-2}

Choose the correct answer from the options given below:

1. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
2. (A)-(II), (B)-(I), (C)-(III), (D)-(IV)
3. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

4. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Options 1. 1

2. 2

3. 3

4. 4

Q.34

If the integral $I = \int \frac{x^2}{\sqrt{1+x}} dx = \frac{1}{\alpha}(1+x)^\alpha - \frac{8\alpha}{15}(1+x)^{\alpha-1} + 2(1+x)^{\alpha-2} + C$,

where C is the constant of integration, then the value of α is:

(1) 1

(2) $\frac{1}{2}$

(3) $\frac{3}{2}$

(4) $\frac{5}{2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.35

If $A = \begin{bmatrix} 0 & a & 2 \\ -2 & 0 & b \\ -2 & 2 & c \end{bmatrix}$ is a skew symmetric matrix, then the value of $(a+b+c)^3$ is:

1. 1

2. 0

3. 8

4. 2

Options 1. 1

2. 2

3. 3

4. 4

Q.36

In a 500 m race, the ratio of speeds of two participants A and B is 4 : 5.

If A has a start of 180 m, then the distance by which A wins is:

1. 100 m

2. 180 m

3. 150 m

4. 120 m

Options 1. 1

2. 2

3. 3

4. 4

Q.37

If the random variable X follows the Poisson distribution such that $P(X = k) = P(X = k + 1)$, then the mean value of X is:

1. k

2. $k - 1$

3. k^2

4. $k + 1$

Options 1. 1

2. 2

3. 3

4. 4

Q.38

Match List-I with List-II

List-I

List-II

(Matrix A)

(Determinant of adjA)

(A) $\begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$

(I) 3

(B) $\begin{bmatrix} 3 & 4 \\ 3 & 6 \end{bmatrix}$

(II) 6

(C) $\begin{bmatrix} 3 & 7 \\ -2 & -4 \end{bmatrix}$

(III) 5

(D) $\begin{bmatrix} 4 & 3 \\ 3 & 3 \end{bmatrix}$

(IV) 2

Choose the correct answer from the options given below:

1. (A)-(II), (B)-(I), (C)-(III), (D)-(IV)
2. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
3. (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
4. (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Which of the following are correct?

- (A) If $a \equiv b \pmod{n}$, then $-a \equiv -b \pmod{n}$
- (B) If $a + b = c$, then $a \pmod{n} + b \pmod{n} \equiv (a + b + c) \pmod{n}$
- (C) If $a \equiv b \pmod{n}$, then $ka \equiv kb \pmod{n}$, $\forall k \in I$
- (D) If $a \equiv b \pmod{n}$, then $(a + k) \equiv (b + k) \pmod{n}$, $\forall k \in I$

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.40

A box contains 10 balls, each marked with one of the digits 0 to 9.
If four balls are drawn successively with replacement from the box,
then the probability that none is marked with the digit 0 is:

1. $\left(\frac{9}{10}\right)^4$
2. $\left(\frac{1}{10}\right)\left(\frac{9}{10}\right)^3$
3. $\left(\frac{1}{10}\right)^4$
4. $\left(\frac{1}{10}\right)^2\left(\frac{9}{10}\right)^2$

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Irregular trends in a time series are caused by

- (A) Epidemics
- (B) Strikes
- (C) Festival season
- (D) Floods

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.42 Pipe A can fill a tank three times faster than pipe B.

If both pipes A and B are running together, they can fill the tank in 12 minutes.

Then the time taken by pipe B alone to fill the tank is:

1. 24 minutes
2. 36 minutes
3. 40 minutes
4. 48 minutes

Options 1. 1

2. 2
3. 3
4. 4

Q.43 If $y = \left(\log \left(x + \sqrt{x^2 + a^2} \right) \right)^2$ and $x \neq \frac{1-a^2}{2}$,

then $(x^2 + a^2) \frac{d^2y}{dx^2} + x \frac{dy}{dx}$ is equal to:

1. 0
2. 1
3. 2
4. 4

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.44 Which of the following are correct?

- (A) A statistical hypothesis is a statement about the population parameter which may be true or false.
- (B) In a statistical hypothesis testing procedure, a Type I error is made when the null hypothesis is accepted when it is false.
- (C) In a statistical hypothesis testing procedure, a Type II error is made when the null hypothesis is rejected when it is true.
- (D) The number of degrees of freedom of a statistic is the number of independent variates used to compute that statistic.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (D) only
3. (A), (B), (C) and (D)
4. (B) and (C) only

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.45 The particular solution of the differential equation $x(1+y^2)dx - y(1+x^2)dy = 0$, given that $y(0) = 1$ is:

1. $xy^2 = x^2 + 1$
2. $y^2 = 2x^2 + 1$
3. $y^2 = 2x^2 + xy + 1$
4. $x^2 = 3y^2 + 2xy + 1$

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.46 If, in a pair of consecutive positive integers, both numbers are greater than 5 and their sum is less than 23, then the number of such pairs is:

1. Two
2. Three
3. Four
4. Five

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.47 A person invested ₹ 2,50,000 in a fund. At the end of the year, the value of the fund is ₹ 3,00,000. If the market price is the same at the end of the year, then the nominal rate of return is:

1. 10%
2. 15%
3. 20%
4. 12%

- Options
1. 1
 2. 2
 3. 3

Q.48 If C is the original value of the asset, S is the scrap value, n is the useful life of the asset and D is the annual depreciation, then

1. $D = \frac{C + S}{n}$
2. $D = \frac{C - n}{S}$
3. $D = n \frac{C}{S}$
4. $D = \frac{C - S}{n}$

Options 1. 1

2. 2
3. 3
4. 4

Q.49 A container contains 100 liters of milk.
From this container, 10 liters of milk is taken out and replaced by water.
This process is repeated two more times.
The amount of milk left in the container now is:

1. 85 liters
2. 70 liters
3. 67.1 liters
4. 72.9 liters

Options 1. 1

2. 2
3. 3
4. 4

Q.50 The point estimate of the population standard deviation for the random sample 5, 8, 10, 7, 10, 14 is:

1. 1.5
2. 4.9
3. 3.1
4. 9

Options 1. 1

2. 2
3. 3
4. 4

Section : Core Mathematics

Q.51 If R and S are two equivalence relations on a set A , then

1. $R \cup S$ is also an equivalence relation.
2. $(R \cup S)^{-1}$ is also an equivalence relation.
3. $(R \cap S)^{-1}$ is also an equivalence relation.
4. $R \cap S$ is not an equivalence relation.

Options 1. 1

2. 2
3. 3
4. 4

Q.52 If $\vec{a}$ is a non-zero vector, then always

1. $\vec{a} \cdot \vec{a} = 0$
2. $\vec{a} \cdot \vec{a} > 0$
3. $\vec{a} \cdot \vec{a} < 0$
4. $\vec{a} \cdot \vec{a} = 1$

Options 1. 1

2. 2
3. 3
4. 4

Q.53 $g(x) = \begin{cases} \alpha x, & x < 0 \\ 5, & x \geq 0 \end{cases}$ is continuous at $x = 0$, then the value of α is

1. 0
2. 5
3. -5

4. any real number

- Options 1. 1
2. 2
3. 3
4. 4

Q.54 The area (sq. units) bounded by the curve $y = \sin x$, $\pi \leq x \leq 2\pi$ and the x -axis is

1. 4
2. 2
3. 1
4. 2π

- Options 1. 1
2. 2
3. 3
4. 4

Q.55 The area enclosed between the graph of $y = x^3$

and the lines $x = 0$, $y = 1$, $y = 8$ is

1. $\frac{45}{7}$ Sq. units
2. $\frac{45}{2}$ Sq. units
3. $\frac{45}{4}$ Sq. units
4. 7 Sq. units

- Options 1. 1
2. 2
3. 3
4. 4

Q.56 Let $A = [a_{ij}]_{n \times n}$ and $B = [b_{ij}]_{n \times n}$.

Then which of the following is/are true?

- (A) $AB = BA$
(B) $(AB)^{-1} = B^{-1}A^{-1}$
(C) $(AB)^T = B^T A^T$
(D) $AB = 0 \Rightarrow A = 0$ or $B = 0$

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (B) and (C) only
3. (B), (C) and (D) only
4. (A) only

- Options 1. 1
2. 2
3. 3
4. 4

Q.57 If $3 \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$, then the values of x, y, z and w are

1. $x = 1, y = 2, z = 3, w = 4$
2. $x = 2, y = 3, z = 1, w = 4$
3. $x = 2, y = 4, z = 1, w = 3$
4. $x = 1, y = 4, z = 2, w = 3$

- Options 1. 1
2. 2
3. 3
4. 4

Q.58 If a line makes angles α, β and γ with positive x -axis, y -axis and z -axis respectively, then the value of

$$\sin^2 \frac{\alpha}{2} \cos^2 \frac{\alpha}{2} + \sin^2 \frac{\beta}{2} \cos^2 \frac{\beta}{2} + \sin^2 \frac{\gamma}{2} \cos^2 \frac{\gamma}{2} \text{ is}$$

1. 1
2. $\frac{1}{2}$
3. 2
4. $\frac{1}{4}$

Options 1. 1

2. 2

3. 3

4. 4

Q.59 Match List-I with List-II

Where $\mathbb{R}$ is set of real numbers

List-I

(A) $f: \mathbb{R} \rightarrow \mathbb{R}$ s.t $f(x) = x^4$ is

(B) $f: \mathbb{R} \rightarrow [0, \infty)$ s.t $f(x) = x^4$ is

(C) $f: [0, \infty) \rightarrow \mathbb{R}$ s.t $f(x) = x^4$ is

(D) $f: [0, \infty) \rightarrow [0, \infty)$ s.t $f(x) = x^4$ is

List-II

(I) one-one, Into

(II) many-one, into

(III) one-one, onto

(IV) many-one, onto

Choose the correct answer from the options given below:

1. (A) – (IV), (B) – (II), (C) – (I), (D) – (III)

2. (A) – (II), (B) – (IV), (C) – (I), (D) – (III)

3. (A) – (II), (B) – (IV), (C) – (III), (D) – (I)

4. (A) – (IV), (B) – (II), (C) – (III), (D) – (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.60 Consider the line $\frac{x-2}{2} = \frac{2y-5}{-3}, z = -1$.

Then which of the following is/are true?

(A) It has Direction ratios (2, -3, -1)

(B) It has Direction cosines $\left(\frac{4}{5}, -\frac{3}{5}, -\frac{1}{5}\right)$

(C) It has Direction ratios $\left(2, -\frac{3}{2}, 0\right)$

(D) It has Direction cosines $\left(\frac{4}{5}, -\frac{3}{5}, 0\right)$

Choose the correct answer from the options given below:

1. (A only)

2. (A and B only)

3. (C and D only)

4. (A and C only)

Options 1. 1

2. 2

3. 3

4. 4

Q.61 A bag contains 4 red and 6 green balls. A ball is drawn at random, its colour is noted and is returned to the bag. One additional ball of the colour drawn is put in the bag. Again a ball is then drawn from the bag. The probability of this ball to be of green colour is

1. $\frac{7}{10}$

2. $\frac{4}{5}$

3. $\frac{3}{5}$

4. $\frac{2}{5}$

Options 1. 1

2. 2

3. 3

4. 4

Q.62 If the points (a_1, b_1) , (a_2, b_2) and (a_1+a_2, b_1+b_2) are collinear, then

1. $a_1b_1 = -a_2b_2$

2. $a_1b_1 = a_2b_2$

3. $a_1b_2 = a_2b_1$

4. $a_1b_2 = -a_2b_1$

Options 1. 1

2. 2

3. 3

4. 4

Q.63 Match \textbf{List-I} with \textbf{List-II}

List-I

List-II

Inverse Trigonometric function Principal values of arguments

(A) $\sin^{-1}\left(-\frac{1}{2}\right)$

(I) $-\frac{\pi}{3}$

(B) $\cos^{-1}\left(-\frac{1}{2}\right)$

(II) $\frac{3\pi}{4}$

(C) $\tan^{-1}(-\sqrt{3})$

(III) $-\frac{\pi}{6}$

(D) $\sec^{-1}(-\sqrt{2})$

(IV) $\frac{2\pi}{3}$

Choose the correct answer from the options given below:

1. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

2. (A) – (III), (B) – (IV), (C) – (II), (D) – (I)

3. (A) – (IV), (B) – (III), (C) – (II), (D) – (I)

4. (A) – (IV), (B) – (III), (C) – (I), (D) – (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.64 A random variable X has the following probability distribution

X 0 1 2 3 4 5

P(X) 0.1 k 0.2 2k 0.3 k

Then P (X < 3) is

1. 0.1

2. 0.2

3. 0.3

4. 0.4

Options 1. 1

2. 2

3. 3

4. 4

Q.65

If $y = e^{a \cos^{-1} x}$, $-1 < x < 1$, then $(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx}$ is equal to

1. $a^2 y$

2. $-a^2 y$

3. ay

4. $-ay$

Options 1. 1

2. 2

3. 3

4. 4

Q.66 Match List-I with List-II

List-I

List-II

\[Functions

Integers

(I) $\log|x + \sqrt{4 + x^2}| + C$

(II) $\sin^{-1}\left(\frac{x}{4}\right) + C$

(III) $\frac{1}{4} \log \left| \frac{x-2}{x+2} \right| + C$

Extra close brace or missing open brace (IV) $\frac{1}{4} \tan^{-1}\left(\frac{x}{4}\right) + C$

Choose the correct answer from the options given below:

1. (A) – (IV), (B) – (III), (C) – (II), (D) – (I)
2. (A) – (III), (B) – (II), (C) – (IV), (D) – (I)
3. (A) – (III), (B) – (IV), (C) – (II), (D) – (I)
4. (A) – (IV), (B) – (III), (C) – (I), (D) – (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.67 If A and B are square matrices of order 3 such that $|A| = -1$ and $|B| = 5$, then the value of $|3AB|$ is

1. – 15
2. – 27
3. – 81
4. – 135

Options 1. 1

2. 2

3. 3

4. 4

Q.68 Given that $\vec{a} = -3\hat{i} - 6\hat{j} + 4\hat{k}$, $\vec{b} = 9\hat{i} - \lambda\hat{j} - 12\hat{k}$

If $\vec{a} \times \vec{b} = \vec{0}$, then the value of λ is

1. – 18
2. 9
3. – 9
4. 18

Options 1. 1

2. 2

3. 3

4. 4

Q.69 $\int_{3\pi/8}^{\pi/8} \frac{\tan^{2025} x}{\tan^{2025} x + \cot^{2025} x} dx$ is equal to

1. $\frac{\pi}{8}$
2. $-\frac{\pi}{8}$
3. $\frac{\pi}{4}$
4. $-\frac{\pi}{4}$

Options 1. 1

2. 2

3. 3

4. 4

Q.70 A box contains 2 black and 4 red balls and another box contains 4 black and 3 red balls. If a ball is drawn at random from one of the two boxes, then the probability of getting a black ball is

1. $\frac{2}{6}$
2. $\frac{4}{7}$
3. $\frac{19}{42}$
4. $\frac{6}{13}$

Options 1. 1

2. 2

3. 3

4. 4

Q.71 Consider the following L.P.P.

Max. $z = 5x + 2y$, subject to $-2x - 3y \leq -6$, $x - 2y \leq 2$, $3x + 2y \leq 12$, $-3x + 2y \leq 3$ and $x, y \geq 0$ then

1. Its feasible region is bounded and has 4 corner points
2. Its feasible region is bounded and has 3 corner points
3. Its feasible region is unbounded and has 4 corner points
4. Its feasible region is unbounded and has 3 corner points

Options 1. 1

2. 2

3. 3

4. 4

Q.72 Match List-I with List-II

List-I

List-II

(A) Angle between $\hat{i}$ and $-\hat{j}$ is

(I) $\frac{\pi}{6}$

(B) Angle between $2\hat{i} + \hat{k}$ and $10\hat{i} + 5\hat{k}$ is (II) $\frac{\pi}{4}$

(C) Angle between $\hat{i}$ and $\hat{i} + \hat{j}$ is

(III) 2π

(D) Angle between $\sqrt{3}\hat{j} - \hat{k}$ and $\hat{j}$ is

(IV) $\frac{\pi}{2}$

Choose the correct answer from the options given below:

1. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

2. (A) – (II), (B) – (IV), (C) – (I), (D) – (I)

3. (A) – (IV), (B) – (III), (C) – (II), (D) – (I)

4. (A) – (IV), (B) – (III), (C) – (I), (D) – (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.73 Given differential equation,

$$(1 + y^2)dx = (\tan^{-1} y - x) dy,$$

then which of the following is/are true?

(A) Integrating factor = $\tan^{-1} x$

(B) Integrating factor = $\tan^{-1} y$

(C) Integrating factor = $e^{\tan^{-1} y}$

(D) Degree = 1

Choose the correct answer from the options given below:

1. (B) and (D) only

2. (A) only

3. (C) only

4. (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.74 In class XII, suppose 5% of boys and 0.25% of girls are physically fit for a game.

A fit student is selected at random from this class having same number of boys and girls.

If the probability that the selected student is a girl is $\frac{m}{n}$,

$\gcd(m, n) = 1$, then $m + n$ is equal to

1. 22

2. 23

3. 24

4. 25

Options 1. 1

2. 2

3. 3

4. 4

Q.75 If the interval in which the function $f(x) = \frac{x}{x^2 + 1}$ is strictly increasing is $(-a, a)$, then a is equal to

1. 1

2. 2

3. 3

4. 4

Options 1. 1

2. 2

3. 3

4. 4

Q.76 Consider the following L.L.P.

Minimize $z = 30x - 30y + 1800$;

subject to $x + y \leq 30$, $x \leq 15$, $y \leq 20$, $x + y \geq 15$ and $x, y \geq 0$.

Then it attains its optimal value at the point

1. (0, 20)

2. (20, 40)

3. (10, 20)

4. (0, 15)

Options 1. 1

2. 2

3. 3

4. 4

Q.77 $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} dx$ is equal to

1. $2 \sin x + 2x \cos \alpha + C$, where C is an arbitrary constant

2. $2 \sin x + 2 \sin \alpha + C$, where C is an arbitrary constant

3. $\frac{\sin 2x}{2} + x \cos 2\alpha + C$, where C is an arbitrary constant

4. $\frac{\sin 2x}{2} + \frac{\sin 2\alpha}{2} + C$, where C is an arbitrary constant

Options 1. 1

2. 2

3. 3

4. 4

Q.78 Match List-I with List-II

Let A be any invertible square matrix. Then

List-I **List-II**

(A) $A - A^T$ (I) $|A|A^{-1}$

(B) AA^T (II) Skew-symmetric

(C) $\det(A^{-1})$ (III) Symmetric

(D) $\text{adj } A$ (IV) $[\det(A)]^{-1}$

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

3. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

4. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.79 If $\vec{a}$ and $\vec{b}$ are two unit vectors and $\vec{a} + \vec{b}$ is also a unit vector, the magnitude of $\vec{a} - \vec{b}$ is

1. 0

2. 1

3. $\sqrt{2}$

4. $\sqrt{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.80 The angle between the pair of lines given by $\vec{r} = \hat{i} + 2\hat{j} - 3\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k})$ and $\vec{r} = 5\hat{i} + \hat{j} + \hat{k} + \mu(3\hat{i} - 2\hat{j} + 6\hat{k})$

1. $\cos^{-1}\left(\frac{21}{19}\right)$
2. $\sin^{-1}\left(\frac{19}{21}\right)$
3. $\cos^{-1}\left(\frac{19}{21}\right)$
4. $\sin^{-1}\left(\frac{21}{19}\right)$

Options 1. 1

2. 2
3. 3
4. 4

Q.81

For $x \in \mathbb{R} - \{0\}$, the function $f(x) = \frac{3}{x} + 7$ is decreasing when

1. $x \in \mathbb{R}$
2. $x \in \mathbb{R} - \{0\}$
3. $x = \{0\}$
4. $x \in \emptyset$; $\emptyset$ is empty set

Options 1. 1

2. 2
3. 3
4. 4

Q.82

The maximum value of $f(x) = \left(\frac{1}{x}\right)^x$ is

1. $e^{-1/e}$
2. $\left(\frac{1}{e}\right)^e$
3. $e^{1/e}$
4. $\left(\frac{1}{e}\right)^{-e}$

Options 1. 1

2. 2
3. 3
4. 4

Q.83

If $\det \begin{pmatrix} 2x & 5 \\ 8 & x \end{pmatrix} = \det \begin{pmatrix} 6 & -2 \\ 1 & 1 \end{pmatrix}$, then the value of x is

1. $\pm 2\sqrt{6}$
2. $\pm 6\sqrt{2}$
3. $\pm \sqrt{6}$
4. $\sqrt{6}$

Options 1. 1

2. 2
3. 3
4. 4

Q.84

The general solution of the differential equation $\frac{dy}{dx} + y \tan x = \sec x$

1. $y \sec x - \tan x = c$, where c is an arbitrary constant
2. $y \sec x + \tan x = c$, where c is an arbitrary constant
3. $y \tan x - \sec x = c$, where c is an arbitrary constant
4. $y \tan x + \sec x = c$, where c is an arbitrary constant

Options 1. 1

2. 2
3. 3
4. 4

Q.85

Match List-I with List-II

\textbf{List-I}

(A) $f(x) = x \cdot \sin x$

(B) $f(x) = \frac{|x|}{x}$, $x \neq 0$ and $f(x) = 1$ at $x = 0$

(C) $f(x) = x - [x]$, $[x]$ denotes greatest integer function (III) is not differentiable at $x = 1$

(D) $f(x) = e^{|x-1|}$ (IV) is not continuous at $x = 0$

\textbf{List-II}

(I) is not continuous at $x = -3$

(II) is continuous everywhere

(III) is not differentiable at $x = 1$

(IV) is not continuous at $x = 0$

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
2. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
3. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
4. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

Options

1. 1
2. 2
3. 3
4. 4